

Micro-vibration suppression methods and key technologies for high-precision space optical instruments

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ABSTRACT

Micro-vibration caused by disturbance sources onboard spacecraft can severely degrade the working environment of high-precision space optical instruments (HPSOIs) in orbit. Some notable micro-vibration suppression methods have been developed in recent decades. Micro-vibration suppression of spacecraft mainly involves three techniques: disturbance source suppression, payload isolation and transfer path suppression. The first two techniques, together forming one of the most practical suppression methods for micro-vibration suppression tasks, are mainly focused on isolation, whereas the third is a novel suppression method developed for the integrated optical satellite (IOS). Two combinations of them have been effectively used in some spacecraft for a number of years. In these three micro-vibration suppression techniques, the inherent structural performance can be adjusted to deal with variations in the space environment. Usually, passive isolation techniques are deployed in aerospace engineering. Active isolators, however, are often proposed to deal with the low-frequency vibration common in spacecraft. Some emerging advanced suppression methods that exploit the benefits of IOS for micro-vibration have also been reported in the literature. This represents an interesting and highly promising approach to solve some challenging problems in the field. This work serves as an effective state-of-the-art review of the micro-vibration suppression methods and/or technologies that have been developed, mainly over the last few years, specifically for or that potentially could be used for micro-vibration suppression onboard spacecraft with HPSOIs.

1. Introduction

With the rapid development of deep space exploration technology, higher requirements for space technology have become necessary. More demanding performance requirements for the resolution, stability and pointing accuracy in spacecraft have increased the requirements for spacecraft payloads, especially those of high-precision space optical instruments (HPSOIs). This is a challenge because HPSOI systems are very sensitive to disturbances, such as vibrations. One of the key technologies that need to be studied to improve HPSOI payloads is micro-vibration suppression technology [1]. The term micro-vibration usually refers to low-level mechanical vibration or disturbance in the microgravity environment, typically occurring at frequencies of less than 1 Hz up to 1 kHz [1–3]. In fact, whether for deep space exploration, high-resolution Earth observation, or inter-satellite measurement, spacecraft micro-vibration is an important factor affecting the pointing

accuracy and imaging quality of high-precision spacecraft [4–7]. Most spacecraft will be affected by micro-vibration. Due to the low amplitude of the micro-vibration dynamic environment, it has no obvious impact on most components of the spacecraft, so it is usually ignored [3]. However, for HPSOI, the micro-vibration environment seriously affects the pointing accuracy and attitude stability of its payloads, and sharply reduces its resolution and other major performance indicators, thus affecting imaging quality. Therefore, the impact of micro-vibration must be considered in the design of HPSOI [6,8,9].

The ground resolution of a high-resolution Earth observation remote sensing satellite is usually between 0.5 and 0.1 m. For instance, KH-12, the most advanced detection satellite of the United States has a resolution of 0.1 m [10], and the resolution of the WorldView-2 is 0.46 m [11]. The ground resolution of SuperView-1 developed by China is 0.5 m [3, 12]. Compared with earth observation satellites, the resolution of a deep space exploration optical spacecraft is usually one or two orders of

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magnitude higher. For example, the angular resolution of Hubble Space Telescope (HST) is 0.1 arc-seconds [13], and the measurement accuracy of the Fine Guide Sensor (FGS) installed onboard HST is 0.3 milli-arc-seconds. The Next-Generation Space Telescope, James Webb Space Telescope (JWST) scheduled to launch in 2021, is expected to reach 4 milli-arc-seconds [14]. NASA, ESA and JAXA have all successfully carried out laser communication experiments between satellite and earth, and also between satellites. In the latest experiment, the distance between satellites is more than 45,000 km, and the relative pointing accuracy among satellites is required to be better than 5 milli-arc-seconds [15].

There are many inducing factors of micro-vibration onboard spacecraft, such as high-speed rotation of rotating parts, step-by-step motion of driving mechanisms of large controllable components, ignition of thruster during orbit change and attitude adjustment, and cold and hot alternation induced disturbance of large flexible structure in and out of the shadow, among which high-speed spinning of rotor represented by reaction flywheel is the typical inducing disturbance factor. Flywheel assemblies are widely used in space technology to provide attitude control and momentum stability of spacecraft. Mostly, the micro-vibration generated by flywheel is divided into two categories [3]: (a) dynamic and static imbalance of flywheel structure and roller structure defects, which are called primary disturbances and (b) internal structure response amplification caused by a primary disturbance of the flywheel. Static imbalance is caused by rotor eccentricity, which is related to rotating speed; dynamic imbalance is caused by shaft instability, and bearing disturbances are caused by irregularities in components such as balls, races, and cage [16].

Dynamic forces and moments generated by a disturbance source can propagate in spacecraft structures, especially the HPSOI. The main micro-vibration of onboard satellites is within the frequency range of 0.1 Hz–300 Hz. Vibration above 300 Hz is classed as high-frequency disturbance, and vibration below 0.1 Hz is classed as low-frequency disturbance. High-frequency disturbance is mainly caused by control momentum wheel assemblies (CMG) and low-frequency disturbance by reaction wheel assemblies. The frequency of a spacecraft attitude control system is generally below 0.1 Hz [15]. Optical image stabilization or image correction can be used to compensate for low-frequency disturbance below 1 Hz. Micro-vibration suppression measures are mainly used to improve the image blurring caused by structural vibration and/or attitude jitter above 1 Hz. Due to rotating machines and their noise, micro-vibration could contain very different frequency components, including harmonic, random, and the harmonic, narrowband frequency is the most common disturbance [17]. With the development of modern spacecraft with the characteristics of low weight, high precision and large span, the influence of micro-vibration onboard spacecraft has become more and more significant [18–21].

Overall, three kinds of suppression techniques can be classified for suppression of micro-vibration onboard spacecraft, for example, disturbance source control, transfer path control and payload control. In this review, these three types of suppression techniques will be discussed, including the suppression mechanism and characteristics, configuration design and features, and advantages and disadvantages. Thereafter, a special section is dedicated to discussing some novel and emerging suppression methods developed by exploring the integrated design of optical remote sensing satellites. This review will serve as a state-of-the-art (although not comprehensive) summary of the micro-vibration suppression methods and/or technologies that have been developed over the last few years.

2. Influence of micro-vibration on optical system of HPSOI

The influence of micro-vibration will be reflected in the optical system of optical spacecraft, involving the rigid body displacement of optical elements such as the reflectors and the changing of the line-of-sight (LOS) [22]. The micro-vibration causes the rigid body

displacement of the optical elements to be accompanied by the change of the surface shape, which affects the modulation transfer function (MTF) of the optical system and leads to the deterioration of the image quality. Additionally, the micro-vibration causes the vibration of the LOS, which makes the optical image surface vibrate relative to the imaging device, and the image quality deteriorates because of the limited imaging exposure time [23].

According to the imaging exposure time, vibration can be classified as either low-frequency LOS vibration or high-frequency LOS vibration. Low-frequency LOS vibration causes LOS drift and LOS jitter, where the former causes image pixels to be repeated or go missing, and the latter causes image smear. High-frequency LOS vibration mainly causes LOS jitter, which makes the image quality worse [24,25]. In Ref. [22], the rigid body displacement of optical elements and the changing of the LOS of an optical system are given, as shown in Figs. 1 and 2, respectively. The rigid body displacement of an optical element is generally defined as the displacement and the rotation relative to its Central Coordinate System (CCS). In the optical design software, the optical surface is defined in the Local Coordinate System (LCS), so the rigid body displacement of the optical surface is defined as the displacement and rotation relative to the LCS. The transformation relationship between the LCS and the CCS is shown in Fig. 1. For LOS jitter, it is necessary to calculate the LOS motion between different sampling exposure times by calculating the Root Mean Square (RMS) spot diameter within the imaging exposure time, that is, the LOS drift, as shown in Fig. 2.

In literature [26], the micro-vibration of a high-precision space telescope with a TDICCD detector is analyzed. The degradation coefficient of MTF caused by surface micro-vibration is studied, as shown in Fig. 3. Regardless of the value of the amplitude equivalent pixel number, m , the overall trend of the MTF is gradually reduced with the increase of the ratio of exposure time to vibration period σ . Although there will be some fluctuations, it can be found by comparing several curves that this fluctuation is obviously caused by the increase of the number of equivalent pixels.

It is a very effective method to build a micro-vibration experimental measurement platform and study the mechanism of micro-vibration on the HPSOI [3,27]. In literature [27], the MTF from the flywheel vibration characteristics to the imaging image shift was constructed, and the simulation results were compared with the test results. The harmonic characteristics of the theoretical and measured image shift results matched well, and they had almost the same typical harmonic factors.

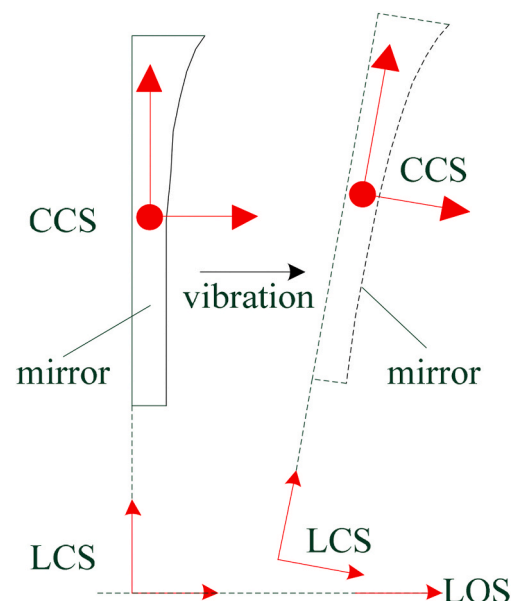


Fig. 1. Diagram of transfer FEA results to optical codes [22].

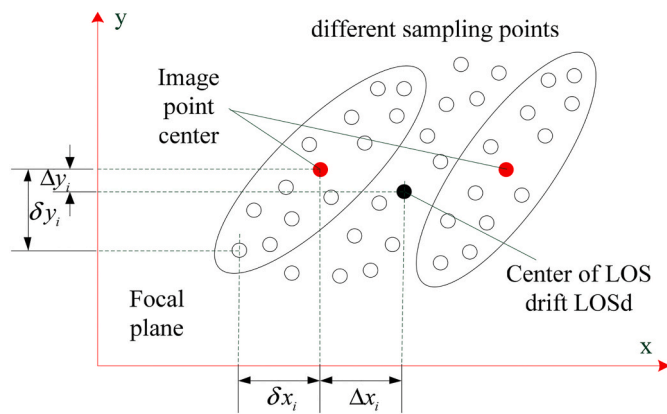


Fig. 2. Diagram of image motion on the focal plane [22].

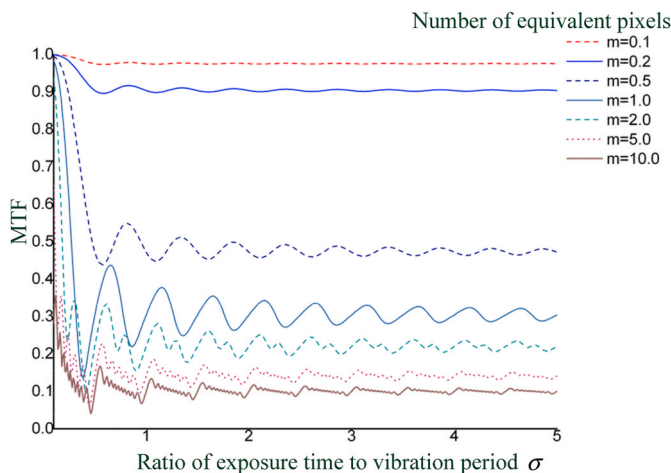


Fig. 3. Diagram of MTF degradation caused by micro-vibration [26].

The modal characteristics had high consistency in the low-frequency range, from the flywheel vibration characteristics to the transfer function expression of image shift. The typical micro-vibration test for an HPSOI is shown in Fig. 4. How to simulate the free boundary conditions of microgravity in orbit is very challenging [3,27–30].

Many micro-vibration suppression methods have been designed to protect high-precision payloads from the impact of micro-vibration onboard spacecraft. Typical isolation systems for HPSOI in spacecraft,

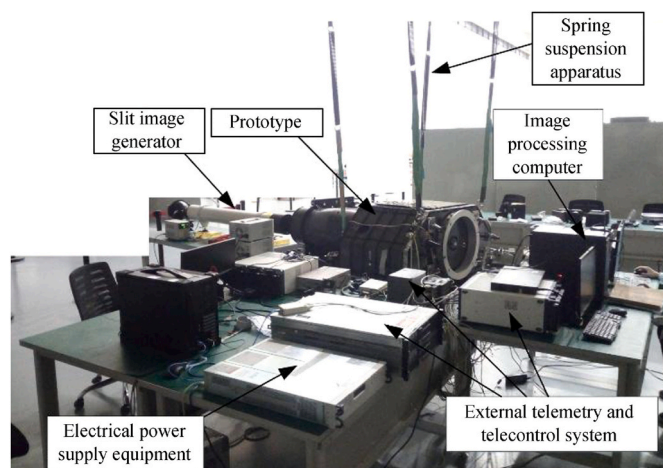


Fig. 4. Photograph of the micro-vibration test bed of an HPSOI [27].

especially space optical telescopes, referred to as vibration isolation and suppression system (VISS) and satellite ultra-quiet isolation technology experiment (SUITE), are described in Refs. [31,32]. The VISS and SUITE, consisting of six struts in a hexapod configuration, are used to support either the HPSOI or the disturbance sources in spacecraft. Many other suppression methods are designed as support structures to suppress micro-vibration transmitted from disturbances of the spacecraft to the HPSOI. The suppression of micro-vibration in the propagation path of disturbances has also been studied.

3. Disturbance source suppression

Micro-vibration suppression technology is currently a hot topic in the field of aerospace engineering. In principle, vibration suppression methods can be divided into two categories: vibration reduction and vibration isolation [3,8,33]. The principle of vibration reduction is to use a damping device to consume the energy of vibration in the position where the structural strain energy is large. The damping device can be made of passive damping material or active piezoelectric ceramic material. The primary principle of vibration isolation is to disconnect the vibration transmission path from the vibration source to the sensitive equipment and install the vibration isolation device between it and the sensitive equipment to prevent the transmission of vibration. The vibration isolation device can be active, passive or an active-passive hybrid [9,34,35]. However, before the suppression of micro-vibration, it is necessary to study its characteristics, which mainly includes two aspects [36,37]: (a) optimize its structure to reduce the disturbance output from the vibration source as much as possible and (b) design the vibration isolation for it.

3.1. Disturbance sources

The moving parts onboard spacecraft may cause micro-vibration, which could change the working environment of the HPSOI. Typical disturbance sources include Reaction Wheel, CMG and scanning mechanism, which have different effects on the HPSOI. Usually, studies focus on the micro-vibration contribution of only the main components of spacecraft. In the following paragraphs, the typical disturbance sources are summarized with a concise and profound description, including the technical diagram and their data analysis.

3.1.1. Reaction wheel

A flywheel assembly is mainly composed of a momentum wheel rotor, bearing, cover and motor. The rotor mass is mostly concentrated on the outer edge of the wheel, providing the control force of the flywheel. The typical cantilever flywheel structure is shown in Fig. 5 [3]. According to the principle of conservation of angular momentum, the flywheel assembly can change the output torque by accelerating or decelerating the rotor driven by the motor, thus changing the attitude of

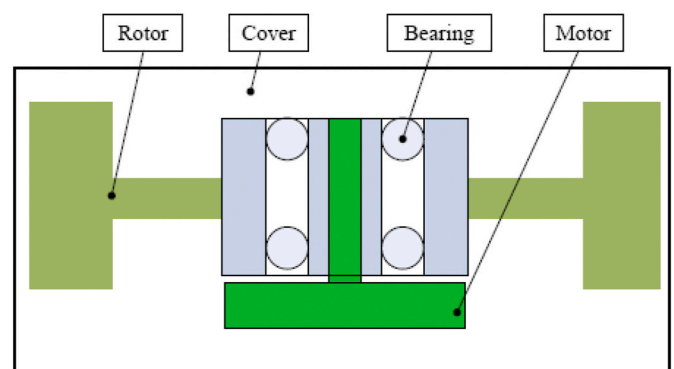


Fig. 5. Structure of cantilever flywheel [3].

the spacecraft [38]. When the flywheel spins normally, it will output a series of disturbing forces and moments besides the expected torque, which are affected by the imbalance of the flywheel, the perfection of the bearing and the defects of the motor. Therefore, the disturbance mechanism of the flywheel can be divided into two mechanisms [6,39]: 1) active disturbance, which is generated by the spinning of the rotor, mainly including the imbalance of the flywheel, the disturbance of the roller bearing and motor, etc., and 2) flywheel structural disturbance, which is the disturbance caused by the internal structural components of the flywheel in response to the active disturbance of the flywheel.

In literature [40], the flywheel assembly used in aerospace engineering is divided into the following four categories: reaction wheel, offset momentum wheel, frame momentum wheel and CMG system. The reaction wheel has zero nominal speed [41,42], and its maximum speed is ± 3600 rpm, which is used to provide control torque for spacecraft attitude. The momentum wheel generally has a high nominal speed of 3000–6000 rpm [42–44] and a large axial moment of inertia. Using the gyroscopic effect, one momentum wheel can keep the satellite passively stable in three axes. At the same time, the momentum wheel can provide an axial control moment [45]. A single frame momentum wheel can realize three-axis stable control of spacecraft attitude in a small angle range [46]. CMG can output large torque, but its volume is also larger [47]. Furthermore, according to whether there is a series supporting system inside the flywheel, the flywheel can be classified as a rigid supporting flywheel or an elastic supporting flywheel, as shown in Fig. 6 [40].

In literature [6], the flywheel components' disturbance mechanism was studied from four aspects, including uneven rotator, rotator friction, bearing disturbance and foundation loose, and a mathematical model of disturbance was built to analyze the characteristics of disturbance. The simulated and tested results of the flywheel components' disturbances were compared and discussed. The disturbance force of flywheel components was around 2 N in magnitude, and the torque of disturbance was around 1.5 N m in magnitude in the time domain. In Ref. [48], a linear and a nonlinear dynamic model were established for the flywheel of spacecraft. The dynamic characteristic of the flywheel was analyzed, and it was found that the soft suspension flywheel can increase the rotational speed and decrease the disturbance.

Because of the complexity of the flywheel structure, the frequency spectrum of flywheel disturbing force/torque has multiple fractional and integral frequency multiplication besides the one-frequency-doubling, which is directly related to the spin speed. When the frequency of the structure coincides with the frequency-doubled component of the spinning speed, the amplitude of the disturbing force is modulated and amplified [49,50]. The typical relationship between disturbing force, spinning speed and frequency is shown in Fig. 7 with a

waterfall plot [27]. The disturbance has a very significant harmonic characteristic, of which the first harmonic amplitude is the most significant [6,16,51]. Generally, the radial rocking mode of the flywheel will be amplified after coupling with the rotor's disturbing force. In the waterfall plot, it often appears in V-shape in other frequency bands [27, 40,49], for example, 400–500 Hz in Fig. 6(b) [27], which does not conform to the harmonic characteristics. The radial rocking mode is also a typical vibration characteristic of the flywheel. Its frequency band is determined by the flexibility of the flywheel and mounting base [1,16]. According to the vibration isolation theory, the more flexible the mounting base is, the lower the frequency and amplitude of the frequency band will be. Different from the harmonic characteristics, the frequency is fixed, and the band is wide and narrow, which can be easily eliminated by vibration isolation. However, the influence of the harmonic characteristics of the flywheel on the HPSOI is often more difficult to address.

3.1.2. CMG assembly

The CMG is used as the actuator of attitude control in modern satellites [52–54]. Because of static and dynamic rotor imbalances and installation errors, the CMG is one of the main vibration sources in spacecraft [40,54–56]. When the CMG generates a control moment, it also generates periodic and broadband disturbance vibration forces, which have a great impact on the pointing precision and stability of the HPSOI [3,55,57]. However, the micro-vibration characteristics of CMGs are not clearly understood [16].

The disturbing mechanism of the CMG is similar to the reaction wheel. However, due to the addition of a low speed shaft to the reaction wheel system, the modulation effect of the CMG structure is more complex than that of the reaction wheel system [58]. Using the CMG mounted on the Z1 truss of the International Space Station (ISS) as an example, the typical structure of a CMG is shown in Fig. 8 [59].

The spinning speed of CMG's rotor is generally much higher than that of the reaction wheel, so the disturbance is generally much higher than that of the reaction wheel. Through the simulation analysis and test of CMG disturbance [60], it is found that when its working speed is 6000 rpm, besides the corresponding 100 Hz disturbance component, there will be 0.6 times harmonic frequency due to bearing defects, so the main disturbance components of CMG are 60 Hz, 100 Hz and their multiple frequencies. In Ref. [37], a large CMG of 70 N m·s belonging to a remote sensing satellite was analyzed, the fundamental frequency of the system as a whole significantly reduced. In Ref. [61], CMG disturbance characteristics due to imbalances and installation errors were analyzed, and the application of the CMG in remote sensing satellites was discussed. The validity of a micro-vibration model was analyzed and tested by numerical simulation, and it was found that CMG interference

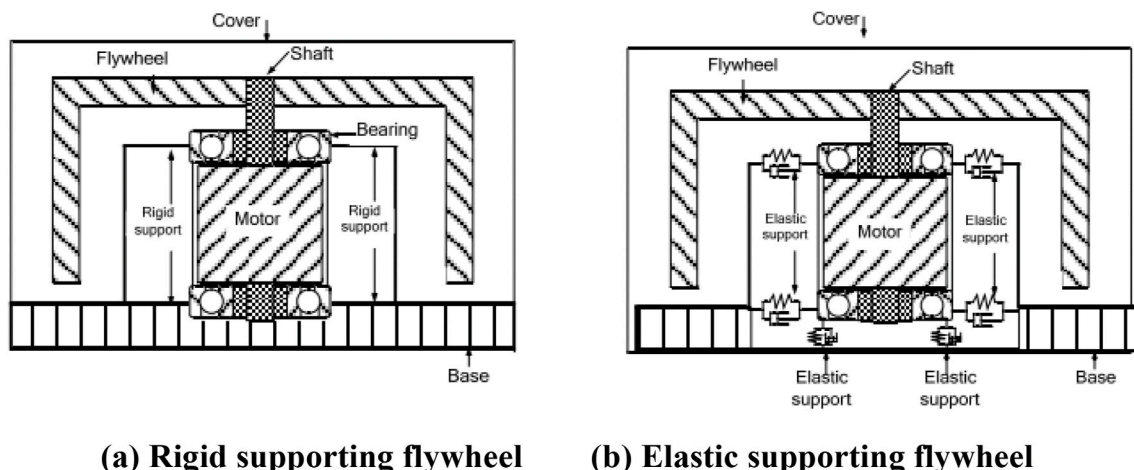


Fig. 6. Diagram of rigid supporting flywheel and elastic supporting flywheel [40].

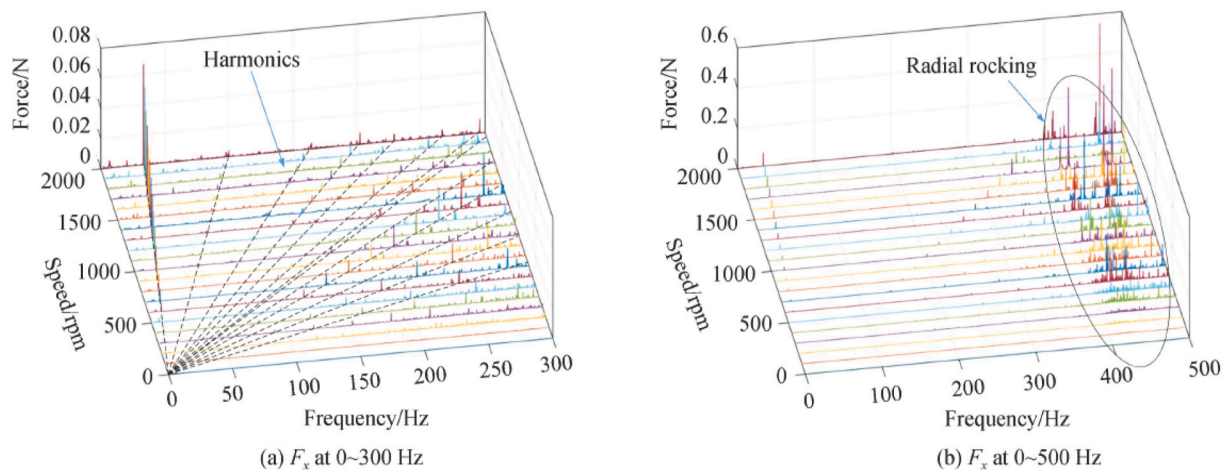


Fig. 7. Waterfall plot of typical wheel disturbing force [27].



Fig. 8. CMGs 1–4 mounted on Z1 Truss of ISS [59].

will significantly affect satellite attitude accuracy and stability. Moreover, the micro-vibration characteristics of single-frame CMG and multi-frame CMG were different. In Ref. [16], a theoretical model describing single-frame CMG was established using an energy method. The model fully considers the flexibility of the internal bearing system and the coupling motion of the flywheel and the universal joint. The micro-vibration characteristics of single frame CMG were tested by using the Kistler vibration table. In Ref. [51], a frequency test under angular

velocity of multi-frame CMG was designed for turbidity removal, and the influence of the frequency from the angle of CMG frames and its spinning speed was calculated, which further provided data for CMG micro-vibration analysis of the spacecraft. Fig. 9 presents the steady-state test vibration disturbance F_x waterfall figure, in which the rated speed of the rotor was 9500 rpm, every 10° of gimbal angle from 0° to 350° was a measuring point, and altogether 36 operation modes were measured.

Disturbance data in orbit is usually displayed by power spectrum density (PSD). The micro-vibration measured onboard a remote-sensing

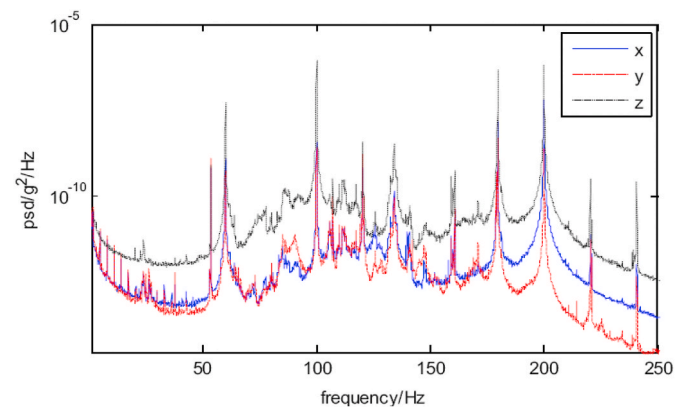


Fig. 10. Acceleration PSD at CMG mounting point [62].

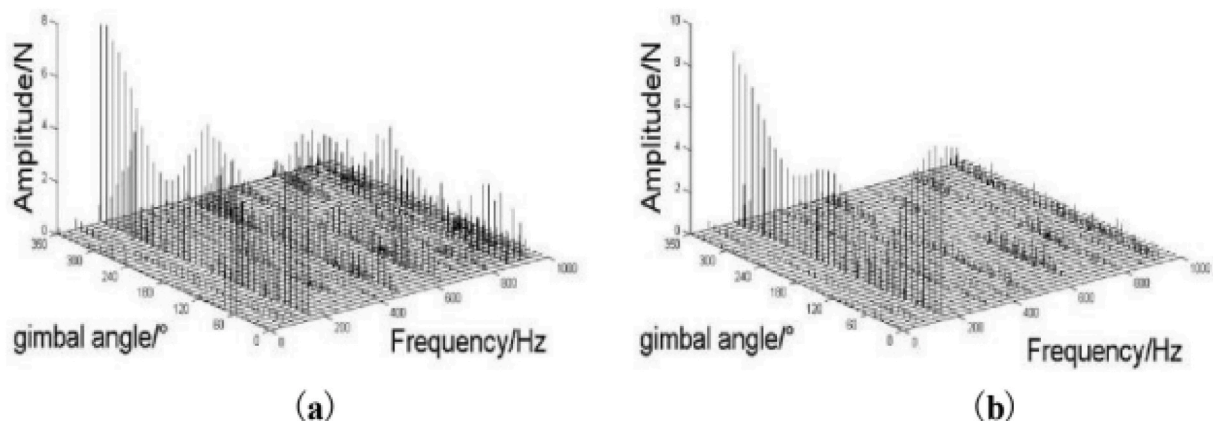


Fig. 9. Steady-state test vibration disturbance F_x waterfall plot [51].

satellite in orbit was presented in Ref. [62]. The disturbance characteristics of CMG are shown in PSD in Fig. 10. It can be seen that the disturbing energy was concentrated at the frequency bands of 60 Hz, 100 Hz, 180 Hz and 200 Hz as four peaks arising after CMG started. This is consistent with the conclusion in Ref. [60].

3.1.3. Scanning mechanism and drive mechanism

Scanning mechanisms and drive mechanisms are also common sources of micro-vibration disturbance in spacecraft. The typical drive mechanism is a solar array drive assembly (SADA) [63–65]. Generally, the sweeping mechanism is driven by a motor to make the swing mirror perform reciprocating motion. Once the swing mirror moves to the limit position, the spring plate is used to limit, buffer and provide reverse torque [66,67]. The continuous reverse movement of the swing mirror produces the continuous pulse reaction torque to the installation foundation. Through the joint action of the leaf spring and the motor, the continuous pulse load is transmitted to the base, causing the micro-vibration of the structure. The spectrum includes not only the disturbance force corresponding to the sweep frequency but also the multiple frequency of sweep frequency. The amplitude of the disturbance force is positively related to the moment of inertia and sweep velocity of the sweep mirror [68]. In Ref. [69], a test system was built to test the micro-vibration characteristics of the fast swing mirror.

The SADA consists of the solar array (SA) drive mechanism to orient the SA wing and to provide power and signal transfer to the spacecraft [64,65]. Fig. 11 shows a diagram providing an overview of SADA [70]. A detailed description of the SADA design can be found in Ref. [71]. The study presented in Ref. [72] analyzed the amplitude of these disturbances of SADA in both time and frequency domains using theoretical methods and experimental techniques. Results acquired in time and frequency domains verify that the increase of subdivisions causes a reduction in micro-vibration generated by SADA. Another study focused on the disturbance properties of SADA driving a flexible load [73]. Result shows that the disturbance frequencies of SADA driving a flexible load consisted of all the natural frequencies of the coupled mechanical system and the input frequency of the electrical pulse signal and its harmonic frequencies. The PSD of the acceleration data acquired in-orbit is shown in Fig. 12. Compared with the figures above, the disturbance of SADA is far lower than other moving parts [62].

3.2. Optimization design of micro-vibration source

The micro-vibration produced by moving parts is mainly caused by

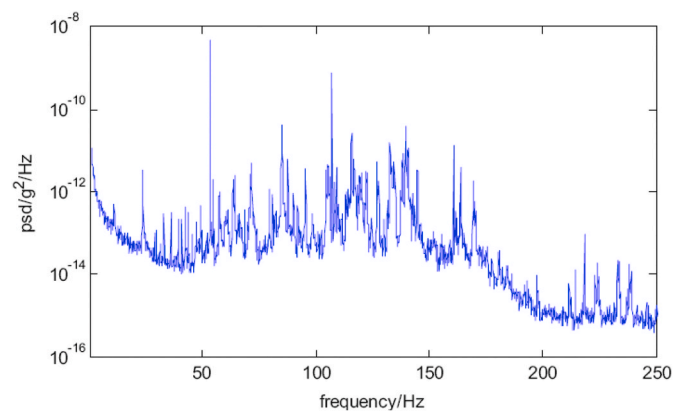


Fig. 12. Acceleration PSD at SADA in orbit [62].

the non-uniform movement of the moving parts [6,55,64]. By optimizing the structure of the vibration source, such as reducing the static and dynamic imbalance of the high-speed rotor, and avoiding the reciprocating mechanism as much as possible, the disturbance of the disturbances can be effectively reduced [74,75].

It is pointed out in Ref. [76] that the performance of flywheel can be optimized by means of structural optimization design to reduce disturbance. Taking the disturbance output response at the interface between the disturbance source and the spacecraft as the optimization target has obvious theoretical and practical significance [77–79]. When the vibration along a certain degree of freedom is decoupled from other degrees of freedom, the energy of external disturbance work is only concentrated on that degree of freedom [80,81]. Hence, by decoupling the six-degree-of-freedom motion of the flywheel system, the complex vibration response of the system in the resonance region can be reduced, and the disturbance performance of the system can be improved [82, 83]. In order to ensure the good performance of the flywheel system in six degrees of freedom, the distribution interval of each natural frequency should be minimized, which will make the natural frequency distribution of the disturbance source reasonable [84]. In Ref. [85], the disturbance model of a spacecraft flywheel and single gimbal CMG system was established, and the disturbance characteristics of the flywheel vibration source were analyzed, which laid an important theoretical foundation for the structural optimization of flywheel systems. Considering the nonlinear effect of the bearing stiffness, the flywheel was simplified as a Jeffcott rotor model, and it was found that a

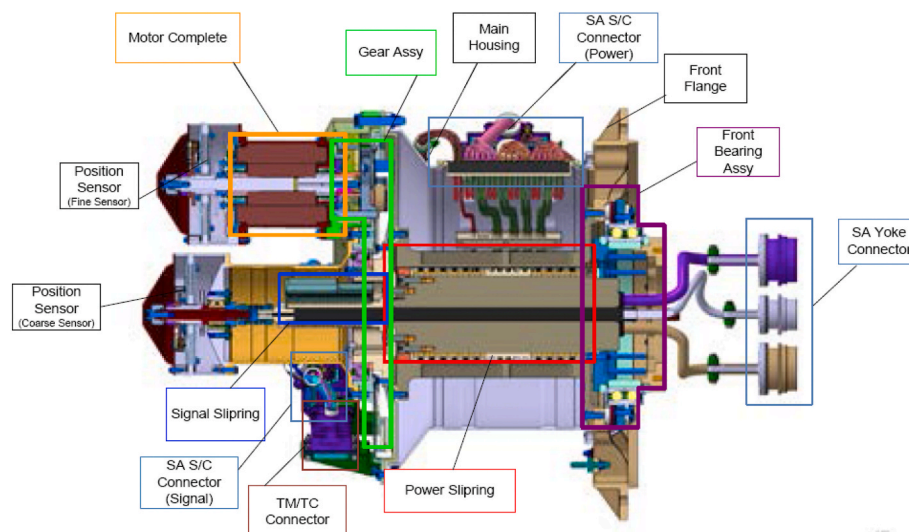


Fig. 11. Diagram of SADA overview [70].

soft suspension can effectively increase the maximum rotational speed and decrease the disturbance force. Based on this, a design method for an elastic-supported flywheel model, as shown in Fig. 13, was proposed to optimize the disturbance source [40]. The sweep mechanism of a space camera was designed as a reciprocating motion of a single swing mirror, and the disturbing force output to the mounting base was in the form of a continuous pulse. After the optimization design, a momentum compensation mechanism was added, which was in inverse motion with the swing mirror. The inertia forces formed by the two mechanisms were offset, and the disturbing force of the output value installation foundation was greatly reduced [15].

3.3. Isolation for micro-vibration source

The early damping device for spacecraft was mainly designed for the harsh dynamic environment during rocket launch [86,87]. The spacecraft micro-vibration suppression device is intended for long-term use in orbit, so its working environment is very different. The micro-vibration amplitude of the dynamic load is at the micrometer level, and it is also subject to special environmental conditions such as space radiation, vacuum and high and low temperature alternation for a long time. Its design requirements are also different from those of previous damping devices. Generally speaking, the micro-vibration damping device of spacecraft should at least meet the following requirements [3,88,89]:

- The damper shall have high sensitivity and provide a high damping ratio at the micrometer level.
- Avoid the movement gap and either minimize the use of or not use motion pairs. In a microgravity space environment, the gap position of the motion pair on the vibration suppression device is in a random state. The micro-vibration generated by the moving parts will cause the collision of the gap surface of the motion pair, resulting in the amplification of the response component.
- The mechanical characteristics of the vibration suppression device cannot be changed obviously under the persistent micro alternating stress. The change of mechanical properties will bring incalculable risks to space photoelectric equipment.
- In the space environment, the vibration suppression device should have good mechanical and chemical stability. That is to say, the stiffness and damping cannot have obvious attenuation, cold welding, or release gas or dust under space radiation, high and low temperature or vacuum environment.

Fig. 14 is the installation diagram of a CMG vibration isolator with all metal material [15]. The metal material has good adaptability to the



Fig. 14. Schematic of CMG vibration isolation [15].

space environment. Through the design of its nonlinear mechanical properties, it has the characteristics of large damping in the launching section and small damping after entering orbit, as well as two functions of resistance environment and micro-vibration isolation in the orbit. In an integrated micro-vibration isolation device proposed in Ref. [1], shown in Fig. 15, four vibration isolator sub-structures are connected as a complete vibration isolator through a connecting plate, and each sub-structure is composed of three bushes, four damping pads and a screw, where the vibration damping pad is made of metal rubber. The attenuation rate of the RMS of the angular displacement was above 50% after installing the vibration isolation device. In Ref. [40], an active control device with elastic support was proposed and shown to effectively suppress the disturbance output of the flywheel.

Honeywell has developed a special flywheel vibration isolator for HST, whose appearance is shown in Fig. 16 [90]. The vibration isolator is a parallel structure of metal spring and low volatility viscous fluid damper, which is mainly used for axial vibration isolation. In the Chandra X-ray observer, a Stewart vibration isolation platform was installed for each flywheel, as shown in Fig. 17 [91,92]. Due to the special requirements on flywheels for the transmission of control torque, and in order to avoid the abnormal vibration of the rotor, the vibration isolation frequency of this kind of vibration isolator cannot be too low, mostly above 15 Hz. The requirements on the stiffness of vibration isolators for momentum wheels and control moment gyroscopes are analyzed in Refs. [93,94], respectively.

In Ref. [9], the vibration isolators used in spacecraft micro-vibration suppression technology in recent years are reviewed in detail, including active vibration isolation, passive vibration isolation and active-passive hybrid vibration isolation. Usually, passive isolation techniques are deployed in aerospace engineering. Active isolators, however, are often proposed to deal with the low frequency vibration that is common in spacecraft. Active-passive hybrid vibration isolation has also been effectively used in some spacecraft structures for a number of years. In semi-active isolation systems, the inherent structural performance can be adjusted to deal with variation in the aerospace environment. This approach is potentially one of the most practical isolation techniques for micro-vibration isolation tasks [3,95,96]. Some emerging advanced vibration isolation methods that exploit the benefits of nonlinearity have also been reported in the literature. This represents an interesting and highly promising approach for solving some challenging problems in the area. It is an effective and less costly method to use vibration isolation to suppress the disturbance caused by micro-vibration sources.

3.4. Problems

Passive vibration isolation methods have the characteristics of highly reliable performance and no energy transmission, which is very satisfactory for micro-vibration suppression. However, characteristics can differ among micro-vibration sources, and the low-frequency vibration

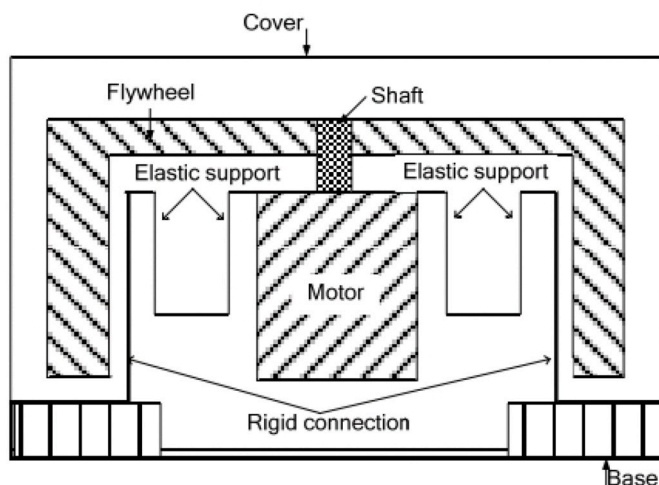


Fig. 13. An elastic-supported flywheel model [40].

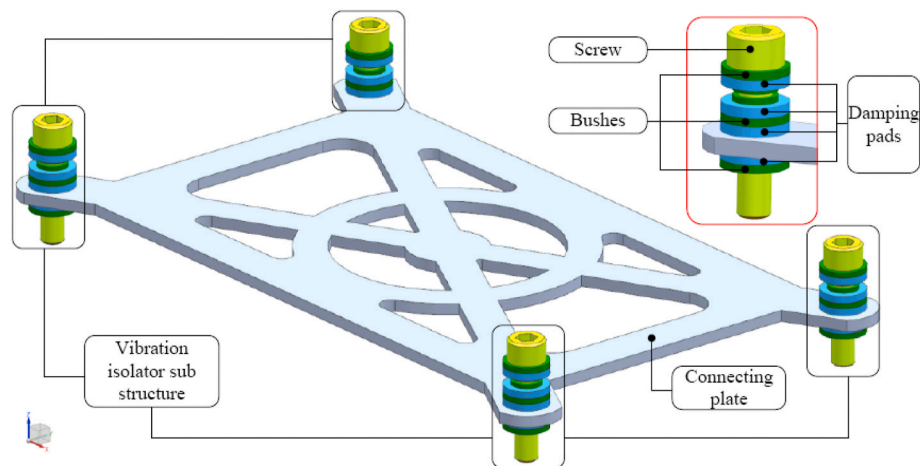


Fig. 15. Artistic model of vibration isolation [1].

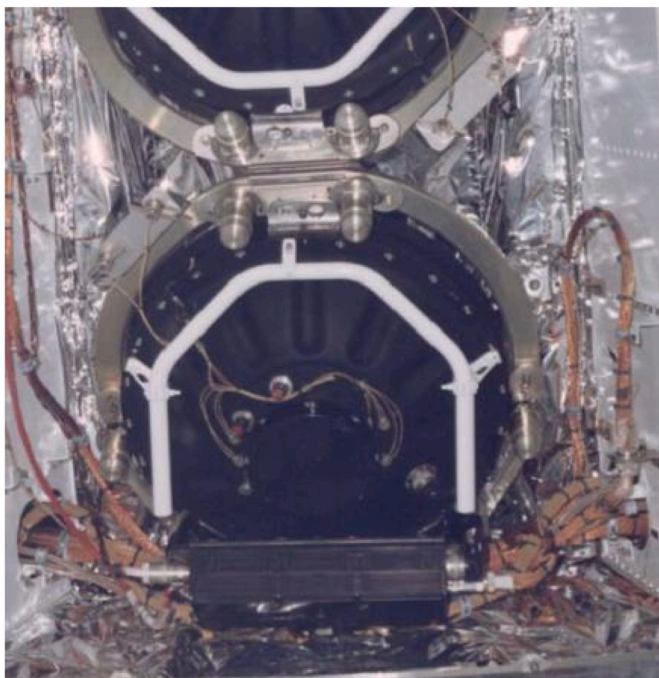


Fig. 16. Momentum wheel isolators onboard HST [90].



Fig. 17. Momentum wheel isolators onboard Chandra X-ray observer [91,92].

suppression effect is often not very good when using pure suppression techniques. Low-frequency disturbance is also very important in spacecraft structural vibration, especially in modern flexible and large-span spacecraft with HPSOI. Therefore, it is still a challenge to employ pure optimization design or source suppression techniques to isolate low-frequency micro-vibration onboard spacecraft.

4. Transfer path suppression

An intelligent control strategy called the transfer path suppression method, which is also known as the change transfer function [3,97,98], is employed in the micro-vibration suppression design of IOS, where self-optimization can be carried out in the change transfer function method. When the output disturbance characteristics of the vibration source are certain, the vibration response of the HPSOI mounting surface on the spacecraft is closely related to the dynamic characteristics of the spacecraft structure. The disturbance from the vibration source to HPSOI in the spacecraft can be effectively reduced by optimizing the dynamic characteristics of the vibration transmission path of the spacecraft structure. This is the novel suppression method developed on IOS in recent years [3,99–105]. A review was presented in Ref. [99] of the performance of a new generation of high-resolution small satellites called Pleiades, which are typical IOSs. The advantages of having an IOS onboard were discussed in depth in Ref. [100]. The integrated design of an optical camera and satellite platform will be a great convenience to the optimization of mechanical environment of the whole satellite [103]. In Ref. [104], the mathematical model of the micro-vibration source of an IOS was established, and the influence of the micro-vibration of the flywheel on the MTF of the HPSOI was analyzed by the integrated modeling method. The micro-vibration environment of IOS is more sensitive, but it is also relatively easy to study [106]. It can be used to establish the influence analysis model of micro-vibration on HPSOI from the disturbance source to the transfer path to the payload, as well as to allocate the index of micro-vibration suppression [3]. This is a novel method of micro-vibration suppression.

Structural dynamics analysis of the spacecraft, made possible by precisely obtaining the structural characteristics of the spacecraft, is the key factor to achieving transfer path suppression [107–111]. According to the modal shape of the spacecraft structure, an effective way to restrain the micro-vibration in the transmission path is to select the reasonable installation location of the vibration source disturbance equipment, avoid the mode antinode of the satellite structure as much as possible and make the installation location as close to the node of the spacecraft structure as possible [108,112]. When the disturbance amplitude transmitted to the payload mounting surface cannot be reduced to an acceptable range through the structural optimization

design or the vibration source location optimization selection, another damping device shall be designed to suppress the micro-vibration in the satellite [3,109].

Usually, the cost of determining the optimal design to suppress the micro-vibration transfer path is high, as it involves the comprehensive satellite/spacecraft structure dynamics. It makes the whole dynamic performance change, which may bring unnecessary trouble to the attitude control system. However, at the beginning of the whole satellite design, the micro-vibration characteristics of the whole platform are analyzed, and effective structural strengthening or restraining measures are taken. The cooperative design of the structural subsystem and control subsystem in the early stage may achieve unexpected results, such as the development period of the whole satellite was shortened to one year, the resolution is in a good level –sub meters, and what is more, the cost is reduced by a third [3,113–115].

5. Payload suppression

5.1. Payload isolation

Usually, payload isolation is realized through the installation of an isolator between the HPSOI and the platform of spacecraft. Many vibration isolation methods have been designed to protect HPSOI from the impact of micro-vibration in spacecraft. Typical isolation systems for space optical telescopes, referred to as VISS and SUITE, are described in Refs. [31,32].

In the structural design of large spacecraft, the coupling of the vibration isolation and the modal shape of large flexible structure should also be considered, and the vibration isolation frequency should not be too low, generally higher than 1 Hz [2]. Tacsat-2 adopts improved D-struts [115] to connect optical load and service cabin for vibration suppression [116,117]. JWST isolated the whole optical payload, that is, the optical load and spacecraft were separated by the vibration isolator (Fig. 18), and the overall vibration isolation frequency reached 1 Hz. In this way, the influence of spacecraft disturbance on the imaging quality of optical load can be minimized. The vibration isolation system is composed of four rod-shaped vibration isolators arranged in a criss-crossed manner. The vibration isolators are made of graphite, and the surface is covered with constrained damping rubber, as shown in Fig. 19 [7].

Based on the six-legged parallel mechanism, the Spacecraft Research and Design Center of the US developed the ultra-quiet platform (UQP) for optical load isolation, which indirectly determines the attitude of the payload by controlling the length of the six legs [118].

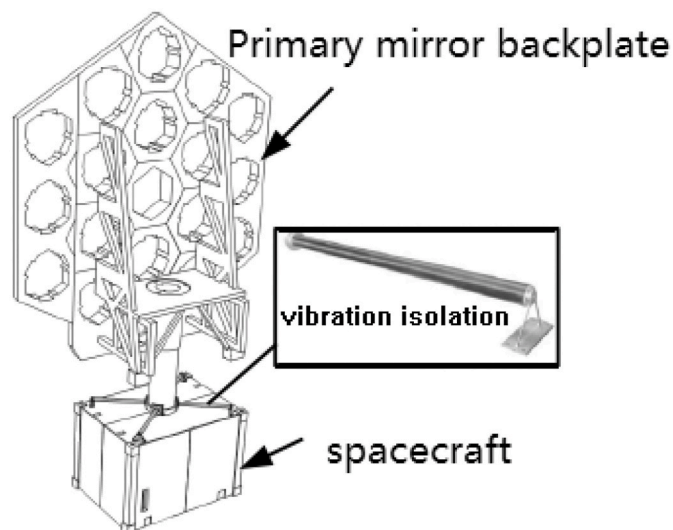


Fig. 18. Micro-vibration isolation onboard JWST.

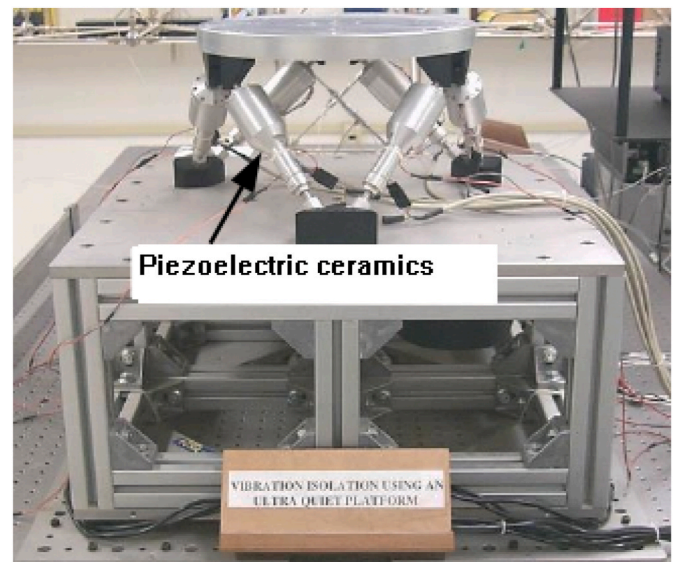


Fig. 19. Ultra-quiet platform [7].

The vibration isolators used in spacecraft micro-vibration suppression technology in recent years are reviewed in detail, including active vibration isolation, passive vibration isolation and active-passive hybrid vibration isolation described in Ref. [9]. The isolation frequency is above 10 Hz, the frequency characteristic of the vibration isolation device is close to the modal frequency of the satellite structure [108], and the modes will be coupled with each other. It is not appropriate to consider the satellite structure as a single rigid body. At this time, the flexibility characteristic of the satellite structure must be taken into account [15]. The design of the vibration isolation should take the flexibility of the whole satellite into account. It is an efficient method to design vibration isolator by adding the damping.

5.2. Reducing payload sensitivity

One effective method to improve the performance of HPSOI in spacecraft is to reduce the sensitivity of the payload and improve the anti-interference ability of the onboard payload. In terms of vibration suppression, this method reduces the micro-vibration response of key elements of the payload.

The most commonly used method to reduce the micro-vibration response of the key elements of the payload is to design and install the vibration isolation device between the camera and the satellite platform, which is similar to the approach of payload isolation, or to adopt a flexible installation form between the sensitive photoelectric conversion element and its supporting structure, so as to reduce the micro-vibration response of the sensitive element and improve the imaging quality [15, 119]. For example, the flexible strut is used to connect the camera's optical mechanism structure and optical elements, which can effectively release the stress, caused by temperature change and installation process, and attenuate the micro-vibration transmitted from the satellite platform structure to the telescope, so as to improve the anti-interference ability of the telescope [120,121].

Generally, optical elements are very sensitive to high frequency vibration and impact [122,123]. Besides the ability of isolating micro-vibration in orbit, the camera vibration isolation device should not deteriorate the mechanical environment of the camera at least. When the stiffness of the on-board damping device is low, the deformation is large under the load condition of the launching section, which may exceed the safety stroke of the damping device. Therefore, protective measures need to be taken for protection. The launching lock is a feasible protective measure. In the launch state, the launch lock is used

to lock the payload and the satellite body, and the launch lock is used to bear the load of the launch section; when in orbit, the launch lock is released, and the vibration isolator is used to connect the payload and the satellite body, so as to isolate the micro-vibration. In essence, the launch lock is a separation device that is used with the vibration isolator.

5.3. Problems

Active-passive hybrid isolation systems combine the best of both the passive and active isolation techniques for a payload suppression strategy. They are effective to isolate the micro-vibration of on-orbit spacecraft for HPSOI. However, implementation of active-passive isolation systems still requires many other accessory devices, such as sensors, actuators, feedback control and power systems, which share the same drawbacks as pure active isolation systems. Semi-active isolation methods could solve some of the problem here, but the most preferable approach would be a novel payload sensitivity reduction method with advantageous suppression performance in practice.

6. Conclusions and discussions

The suppression of micro-vibration onboard spacecraft is a challenging and hot topic to the further development of space engineering. Due to very tiny environmental damping in aerospace, micro-vibration could persist for a long time, which would severely degrade the working environment of sensitive payloads. Three major spacecraft micro vibration suppression technologies have been developed and applied for a long time, including disturbance source control, payload control and transfer path control.

Among the three suppression techniques above, passive, active, active-passive hybrid and semi-active isolation methods are mostly used in source suppression and payload isolation. Passive isolation techniques are commonly used for their high reliability and low cost both in development and energy. However, the micro-vibration in the low frequency region cannot be well attenuated with traditional pure passive isolators. Therefore, novel structures or materials are required for better isolation performance such as those emerging methods exploring structure optimization design benefits in micro-vibration suppression. Active isolation techniques can be effectively used to suppress micro-vibration onboard spacecraft, especially in the low-frequency range if energy cost and weight increase are acceptable. The active damping and active anti-phase control techniques are mainly employed for spacecraft micro-vibration control.

Therefore, it is still a challenge to employ pure structure optimization design or source suppression techniques to control low-frequency micro-vibration. Usually, the cost of identifying the optimal design of the micro-vibration transfer path is high, since it involves the overall satellite structure dynamics. Nonetheless, the cooperative design of the structural and control subsystem in the whole period of satellite design will greatly shorten the development time and cost. Semi-active isolation methods could solve some of the problems here, but the most preferable approach would be novel method to reduce payload sensitivity with advantageous suppression performance in practice.

The optimization design for control micro-vibration sources and reducing payload sensitivity can be used in combination with novel structure systems to achieve an advantageous micro-vibration suppression performance, and thus, could be of much greater significance to the general field of micro-vibration suppression. Noticeably, some emerging advanced Disturbance-Free Payload satellite that exploits the benefits of non-contact actuators is the shape of things to come.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] L. Li, L. Tan, L. Kong, et al., The influence of flywheel micro-vibration on space camera and vibration suppression, *Mech. Syst. Signal Process.* 100 (2018) 360–370.
- [2] G.S. Aglietti, R.S. Langley, E. Rogers, et al., Model building and verification for active control of microvibrations with probabilistic assessment of the effects of uncertainties, *Proc. IME C J. Mech. Eng. Sci.* 218 (4) (2004) 389–399.
- [3] L. Li, Flywheel Micro-vibration Mechanism and Suppression Integrated Study for High Resolution Optical Satellite, University of Chinese Academy of Sciences (Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences), 2018.
- [4] Y. Kong, H. Huang, Vibration isolation and dual-stage actuation pointing system for space precision payloads, *Acta Astronaut.* 143 (2018) 183–192.
- [5] F. Angeletti, P. Gasbarri, M. Sabatini, Optimal design and robust analysis of a net of active devices for micro-vibration control of an on-orbit large space antenna, *Acta Astronaut.* 164 (2019) 241–253.
- [6] L. Li, L. Tan, L. Kong, et al., Flywheel micro-vibration characters of a high resolution optical satellite, *Journal of Vibroengineering* 19 (6) (2017) 3981–3993.
- [7] A.J. Bronowicki, Vibration isolator for large space telescopes, *J. Spacecraft Rockets* 43 (1) (2006) 45–53.
- [8] D.K. Kim, Micro-vibration model and parameter estimation method of a reaction wheel assembly, *J. Sound Vib.* 333 (18) (2014) 4214–4231.
- [9] C. Liu, X. Jing, S. Daley, et al., Recent advances in micro-vibration isolation, *Mech. Syst. Signal Process.* 56 (2015) 55–80.
- [10] A. Mazhar, F. Hassan, M.R. Anjum, et al., High resolution image processing for remote sensing application, in: 2016 Sixth International Conference on Innovative Computing Technology (INTECH), IEEE, 2016, pp. 302–305.
- [11] H. Andri, N. Sella, P. Alfita, et al., Incremental improvement of high resolution satellite imagery for participatory mapping in land registration, in: IOP Conference Series: Earth and Environmental Science, vol. 280, IOP Publishing, 2019, 012035, 1.
- [12] X. Lv, D. Ming, Y.Y. Chen, et al., Very high resolution remote sensing image classification with SEEDS-CNN and scale effect analysis for superpixel CNN classification, *Int. J. Rem. Sens.* 40 (2) (2019) 506–531.
- [13] C. Baranec, R. Riddle, N.M. Law, et al., Bringing the visible universe into focus with Robo-AO, *JoVE* (72) (2013), e50021.
- [14] C. Starr, The James Webb Space Telescope Presentation to SEDS, 2019.
- [15] Qing jun Zhang, Guang yuan Wang, Gang tie Zheng, Micro vibration attenuation methods and key techniques for optical remote sensing satellite, *Journal of Astronautics* 36 (2) (2015) 125–132.
- [16] Q. Luo, D. Li, W. Zhou, et al., Dynamic modelling and observation of micro-vibrations generated by a single gimbal control moment gyro, *J. Sound Vib.* 332 (19) (2013) 4496–4516.
- [17] Vaillon, L., Philippe, C. Passive and active microvibration control for very high pointing accuracy space systems[J]. *Smart Mater. Struct.*, 8(6):719-728.
- [18] Y. Zhang, J. Jiang, G. Zhang, et al., High-accuracy location algorithm of planetary centers for spacecraft autonomous optical navigation, *Acta Astronaut.* 161 (2019) 542–551.
- [19] J. Yang, Z. Xu, Q. Wu, et al., Design of a vibration isolation system for the space telescope, *J. Guid. Contr. Dynam.* 38 (12) (2015) 2441–2448.
- [20] Y. Chen, H. Wen, D. Jin, Design and experiment of a noncontact electromagnetic vibration isolator with controllable stiffness, *Acta Astronaut.* 168 (2020) 130–137.
- [21] D. Yu, M. Lian, F. Zhou, et al., Influence of micro-vibration on the image quality of a GEO remote sensor, *SCIENTIA SINICA Informationis* 49 (1) (2019) 74–86.
- [22] Zhu Jun-qing, Sha Wei, Chen Chang-zheng, et al., Frequency response of imaging quality by micro-vibration for large-aperture space-borne telescope, *Optic Precis. Eng.* 24 (5) (2016) 1118–1127.
- [23] Hai-long Liu, Cheng-shan Han, Li Xiang-zhi, et al., Vibration parameter measurement of TDICCD space camera with mechanical assembly, *Optic Precis. Eng.* 23 (3) (2015) 729–737.
- [24] V. Capuano, E. Shehaj, P. Blunt, et al., High accuracy GNSS based navigation in GEO, *Acta Astronaut.* 136 (2017) 332–341.
- [25] V.L. Genberg, G.J. Michels, K.B. Doyle, Integrated modeling of jitter MTF due to random loads, *Proceedings of SPIE - The International Society for Optical Engineering* 8127 (3) (2011) 777–785.
- [26] Xu Bo-qian, Study on Image Compensation Technology for Spaceborne Cameras under Micro-vibration Circumstances, University of Chinese Academy of Sciences (Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences), 2015.
- [27] Gong Xiao-Xue, Lei Zhang, Xuan Ming, Modeling and ground-based experimental jitter researches on earth observation satellite, *Acta Photonica Sin.* 48 (3) (2019), 0312004.

- [28] R. Benvenuto, M. Lavagna, P. Lunghi, et al., Tethered-tugs for active debris removal: microgravity experimental validation of dynamics and control, in: 7th European Conference on Space Debris, ESA/ESOC. ESA, 2017, pp. 1–11.
- [29] R. Friedman, D.L. Urban, Contributions of Microgravity Test Results to the Design of Spacecraft Fire Safety Systems, 1993.
- [30] J.L. Robey, NASA's international microgravity strategic planning for the space station era, *Acta Astronaut.* 47 (2–9) (2000) 599–606.
- [31] R. Erwin, V. Babuska, L. Sullivan, et al., On-Orbit Active Vibration Isolation: the Satellite Ultraquiet Isolation Technologies Experiment (SUITE), Aiaa Space Conference & Exposition, 2013.
- [32] R.G. Cobb, J.M. Sullivan, A. Das, et al., Vibration isolation and suppression system for precision payloads in space, *Smart Mater. Struct.* 8 (6) (1999) 798–812.
- [33] X. Wang, X. Yue, H. Dai, et al., Vibration suppression for post-capture spacecraft via a novel bio-inspired Stewart isolation system, *Acta Astronaut.* 168 (2020) 1–22.
- [34] Q. Yuan, Y. Liu, N. Qi, Active vibration suppression for maneuvering spacecraft with high flexible appendages, *Acta Astronaut.* 139 (2017) 512–520.
- [35] J.P. Henderson, A. Plummer, N. Johnston, An electro-hydrostatic actuator for hybrid active-passive vibration isolation, *International Journal of Hydromechanics* 1 (1) (2018) 47–71.
- [36] Christopher Emil Eyerman, A System Engineering Approach to Disturbance Minimization for Spacecraft Utilizing Controlled Structures technology, Master Degree, MIT, 1990.
- [37] C. Sheng, Y. Zhang, Q. Hu, et al., Configuration design and collision avoidance control of an electromagnetic vibration isolation system on satellites considering environmental disturbances, *Acta Astronaut.* 165 (2019) 229–241.
- [38] Y.N. Yang Zhao, Y. Y. Earth-observation satellite attitude control using passive and active hybrid magnetically suspended flywheels, *J. Beijing Inst. Technol. (Soc. Sci. Ed.)* 2 (2014) 226–234.
- [39] S. Adhikari, R. Karki, Integrated disturbance response modelling to quantify the operational reliability benefits OF flywheel energy storage systems, *OPERATING RISK ASSESSMENT OF MODERN POWER SYSTEM IN PRESENCE OF FLYWHEEL ENERGY STORAGE* (2018) 38.
- [40] Weiyong Zhou, Research on Dynamic Modeling and Vibration Control for the Flywheel of Spacecraft, National University of Defense Technology, 2012.
- [41] J. Kampmeier, R. Larsen, L.F. Miglionini, et al., Reaction wheel performance characterization using the kepler spacecraft as a case study, in: 2018 SpaceOps Conference, 2018, p. 2563.
- [42] M. Sugita, Torque distribution algorithm for effective use of reaction wheel torques and angular momentums, *Acta Astronaut.* 139 (2017) 18–23.
- [43] Z. Zhang, W.Y. Zhou, W.G.S. Aglietti, G. Bianchi, Modelling microvibrations emitted by satellites reaction wheel assemblies, in: 10th International Conference on Recent Advances in Structural Dynamics, Southampton, UK, 12–14 Jul 2010.
- [44] K. Sathyan, H.Y. Hsu, S.H. Lee, K. Gopinath, Long-term lubrication of momentum wheels used in spacecrafts-an overview, *Tribol. Int.* (43) (2010) 259–267.
- [45] Jingbo Wei, Control Technology of the Momentum Wheel of Maglev Micro Frame, National University of Defense Technology, 2016.
- [46] X. Roser, M. Sghedoni, Control Moment Gyroscopes (CMG's) and Their Application in Future Scientific Missions[J].
- [47] Yoon, Hyungjoo, Tsiotras, Panagiotis. Spacecraft adaptive attitude and power tracking with variable speed control moment gyroscopes[J]. *J. Guid. Contr. Dynam.*, 25(6):1081–1090.
- [48] Q. Luo, D. Li, W. Zhou, Studies on vibration isolation for a multiple flywheel system in variable configurations, *J. Vib. Contr.* 21 (1) (2015) 105–123.
- [49] Z. Zhang, G.S. Aglietti, W. Ren, Coupled microvibration analysis of a reaction wheel assembly including gyroscopic effects in its acceleration, *J. Sound Vib.* 332 (22) (2013) 5748–5765.
- [50] R. Varatharajoo, A combined energy and attitude control system for small satellites, *Acta Astronaut.* 54 (10) (2004) 701–712.
- [51] X. Li, W. Cheng, Test and analysis of micro-vibrations generated by large control moment gyroscope, *International Journal of Control and Automation* 7 (2) (2014) 297–308.
- [52] W. MacKunis, F. Leve, P.M. Patre, et al., Adaptive neural network-based satellite attitude control in the presence of CMG uncertainty, *Aero. Sci. Technol.* 54 (2016) 218–228.
- [53] Z. Jingrui, J. Jin, L. Zaozhen, An improved installation for control moment gyros and its applications on reconfiguration and singular escape, *Acta Astronaut.* 85 (2013) 93–99.
- [54] B.J. Kawak, Development of a low-cost, low micro-vibration CMG for small agile satellite applications, *Acta Astronaut.* 131 (2017) 113–122.
- [55] Y. Zhang, J. Zhang, Disturbance characteristics analysis of CMG due to imbalances and installation errors, *IEEE Trans. Aero. Electron. Syst.* 50 (2) (2014) 1017–1026.
- [56] T. Liang, C. Yi-qing, Model development and adaptive imbalance vibration control of magnetic suspended system, *Acta Astronaut.* 65 (9–10) (2009) 1506–1514.
- [57] P.Y. Bely, Pointing and control, The Design and Construction of Large Optical Telescopes (2003) 252–310.
- [58] C. Mittelsteadt, E. Mehiel, Cal Poly spacecraft attitude dynamics simulator: CP/sads, in: AIAA Guidance, Navigation and Control Conference and Exhibit, 2007, p. 6443.
- [59] C. Gurriss, R. Seidel, S. Dickerson, et al., Space Station Control Moment Gyroscope Lessons Learned, 2010.
- [60] Ruizhi Luo, Gang Hu, Quanwu Wang, Axial Vibration analysis and validity of high-speed rotor of cantilever SGCMG based on a triangle-star equivalent transformation, *Aerospace Control and Application* 37 (2) (2011) 14–20.
- [61] Y.A.O. Zhang, J. Zhang, Disturbance characteristics analysis of CMG due to imbalances and installation errors, *IEEE Trans. Aero. Electron. Syst.* 50 (2) (2014) 1017.
- [62] D. Yu, G. Wang, Y. Zhao, On-orbit measurement and analysis of the micro-vibration in a remote-sensing satellite, *Advances in Astronautics Science and Technology* 1 (2) (2018) 191–195.
- [63] S.E. Woodard, R.R. Lay, R.F. Jarnot, et al., Experimental investigation of spacecraft in-flight disturbances and dynamic response, *J. Spacecraft Rockets* 34 (2) (1997) 199–204.
- [64] R. Li, F. Guo, C. Yu, et al., Development and validation of a mechatronic solar array drive assembly for mini/micro-satellites, *Acta Astronaut.* 134 (2017) 54–64.
- [65] P.K. Maharana, P.S. Goel, Simulation of solar array slewing of Indian remote sensing satellite, *Acta Astronaut.* 25 (3) (1991) 157–164.
- [66] M. Sattar, C. Wei, R. Sattar, Micro vibrations behavior investigation of Solar Array Drive Assembly operating under different subdivisions and loadings, in: 2017 14th International Bhurban Conference on Applied Sciences and Technology (IBCAST), IEEE, 2017, pp. 50–57.
- [67] Q. Yong, M.A.N. Shun-qiang, Influence analysis of spacecraft's attitude based on variable-structure control to decrease motion of scanning camera, *Aerospace Shanghai* 29 (6) (2012), 7–10+15.
- [68] R.G. White, R.J. Pinnington, Practical application of the rapid frequency sweep technique for structural frequency response measurement, *Aeronaut. J.* 86 (855) (1982) 179–199.
- [69] Chuanxin Wei, Research on Kinetics of the Image Stabilization System in Space Telescope, University of Chinese Academy of Sciences (Shanghai Institute of Technical Physics of the Chinese Academy of Sciences), 2016.
- [70] J. Zemann, B. Heinrich, A. Skulicz, et al., Risk mitigation testing with the bepi colombo mpo SADA, in: 15th European Space Mechanisms and Tribology Symposium, 2013, p. 718.
- [71] B. Heinrich, J. Zemann, F. Rottmeier, Development of the BepiColombo MPO solar array drive assembly (SADA), *ESMATS* 2011 (2011).
- [72] J. Zemann, B. Heinrich, A. Skulicz, et al., Risk mitigation testing with the bepi colombo mpo SADA, in: 15th European Space Mechanisms and Tribology Symposium, 2013, p. 718.
- [73] J.P. Chen, W. Cheng, Y.F. Wang, Modeling and simulation of solar array drive assembly disturbance driving a flexible load, in: *Applied Mechanics and Materials*, vol. 565, Trans Tech Publications Ltd, 2014, pp. 67–73.
- [74] L. Qiang, W. Kun, R. Yuan, et al., Optimization design of launch locking protective device (LLPD) based on carbon fiber bracket for magnetically suspended flywheel (MSFW), *Acta Astronaut.* 154 (2019) 9–17.
- [75] B.L. Kantsiper, J.C. Ray, J.W. Hunt, et al., Autonomous avoidance of structural resonances on the STEREO mission, in: *Proceedings of AIAA Guidance, Navigation and Control Conference and Exhibit*, AIAA, Hilton Head, 2007.
- [76] R. Grandhi, Structural optimization with frequency constraints—a review, *AIAA J.* 31 (12) (1993) 2296–2303.
- [77] A. Aghalari, M. Shahravi, Nonlinear electromechanical modelling and dynamical behavior analysis of a satellite reaction wheel, *Acta Astronaut.* 141 (2017) 143–157.
- [78] J.S. Tao, G.R. Liu, K.Y. Lam, Design optimization of marine engine—mount system, *J. Sound Vib.* 235 (3) (2000) 477–494.
- [79] S. Xie, S. Or, H. Chan, et al., Design optimization of machinery mounting systems with an elastic support structure, *Eng. Optim.* 39 (2) (2007) 229–244.
- [80] J.S. Sui, C. Hoppe, J. Hirshey, Powertrain Mounting Design Principles to Achieve Optimum Vibration Isolation with Demonstration Tools, *SAE Technical Paper*, 2003.
- [81] L. Zhenhua, L.J.F. Ranglin, Design and analysis method for improving vibration isolation performance of automotive powerplant mounting systems, *China Mech. Eng.* (3) (2003) 26.
- [82] Y. Zheng, Z. Zhou, H. Huang, A multi-frequency MIMO control method for the 6DOF micro-vibration exciting system, *Acta Astronaut.* (2020).
- [83] R. Naseri, A. Ohadi, V. Fakhari, et al., Optimal characteristics determination of engine mounting system using TRA mode decoupling with emphasis on frequency responses, *Journal of Theoretical and Applied Vibration and Acoustics* 3 (2) (2017) 111–126.
- [84] A. Preumont, *Vibration Control of Active Structures: an Introduction*, third ed., Springer-Verlag Berlin Heidelberg, 2011.
- [85] Qing Luo, Researches on Micro-vibration Characteristics and Isolation Methods of Spacecraft Flywheel System, National University of Defense Technology, 2014.
- [86] G. Matthews, K. Havey Jr., R. Egerman, A paradigm shift to enable more cost effective space science telescope missions in the upcoming decades, in: *Proceedings of Modeling, Systems Engineering, and Project Management for Astronomy IV*, SPIE, San Diego, 2010.
- [87] Y. Zheng, Z. Zhou, H. Huang, A multi-frequency MIMO control method for the 6DOF micro-vibration exciting system, *Acta Astronaut.* (2020).
- [88] M.B. McMickell, T. Kreider, E. Hansen, et al., Optical payload isolation using the miniature vibration isolation system, in: *Proceedings of Industrial and Commercial Applications of Smart Structures Technologies 2007*, SPIE, San Diego, 2007.
- [89] T. Shimizu, S. Nagata, S. Tsuneta, et al., Image stabilization system for Hinode (Solar-B) solar optical telescope, *Sol. Phys.* 249 (221–232) (2008).
- [90] B. Agrawal, Jitter control for imaging spacecraft, in: *Recent Advances in Space Technologies*, 2009. RAST'09. 4th International Conference, 2009, pp. 615–620.

- [91] P. Brugarolas, J. Trauger J. Alexander, et al., ACCESS pointing control system, SPIE Space Telescopes and Instrumentation 2010: Optical, Infrared, and Millimeter Wave 7731 (2010) 1–22.
- [92] Y. Zhang, Z. Guo, H. He, et al., A novel vibration isolation system for reaction wheel on space telescopes, *Acta Astronaut.* 102 (2014) 1–13.
- [93] C. Wu, X. Kong, Y. Liu, et al., Coupling characteristics analysis for the disturbance free payload spacecraft, *Acta Astronaut.* 138 (2017) 407–416.
- [94] Y. Zhang, J.R. Zhang, Analysis of influence of vibration isolation platform on attitude control system and its parameter selection, *Journal of Astronautics* 34 (5) (2013) 657–664.
- [95] D. Kamesh, R. Pandiyan, A. Ghosal, Modeling, design and analysis of low frequency platform for attenuating micro-vibration in spacecraft, *J. Sound Vib.* 329 (2010) 3431–3450.
- [96] Viet Duc La, My Thi Tra Phan, Optimization of two-tuning-knob single-sensor strategy for semi-active isolation, *J. Sound Vib.* 434 (2018) 126–143.
- [97] K. Hoshijima, M. Ikeda, Vibration suppression control for mechanical transfer systems by jerk reduction, *Int. J. Contr. Autom. Syst.* 5 (6) (2007) 614–620.
- [98] K. Togai, K. Choi, T. Takeuchi, Vibration suppression strategy with model based command shaping: application to passenger car powertrain, in: Proceedings of the 41st SICE Annual Conference. SICE 2002, vol. 2, IEEE, 2002, pp. 941–943.
- [99] Wei Xu, yongjie Piao, Analysis of new generation high-performance small satellite technology based on the Pleiades, *Chinese Optics* 6 (1) (2013) 9–19.
- [100] T. Riel, A. Sinn, C. Schwaer, et al., Iterative trajectory learning for highly accurate optical satellite tracking systems, *Acta Astronaut.* 164 (2019) 121–129.
- [101] Ludwig K P, Lawson R R. Payload Container and Fastening Structure for Use in Space Vehicles: U.S. Patent vol. 4,771,971[P]. 1988-9-20.
- [102] E. Cianca, R. Prasad, M. De Sanctis, et al., Integrated satellite-HAP systems, *IEEE Commun. Mag.* 43 (12) (2005) suppl. 33–suppl. 39.
- [103] Luyang Tan, Design and Dynamic Optimization on Integrated Structure of Satellite Main Bearing Structure and Optical camera[D], University of Chinese Academy of Sciences (Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences), 2017.
- [104] Changcheng Deng, Effects of the Flywheel Micro-vibration on the Image Quality of a Integration Satellite, University of Chinese Academy of Sciences (Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences), 2017.
- [105] M.T. Knopp, A. Spoerl, M. Gnat, et al., Towards the utilization of optical ground-to-space links for low earth orbiting spacecraft, *Acta Astronaut.* 166 (2020) 147–155.
- [106] Li Lin, Wand Dong, Jing Xu, et al., Influence of micro-vibration of flywheel components on optical axis of high resolution optical satellite, *Editorial Office of Optics and Precision Engineeri* 24 (10) (2016) 2515–2522.
- [107] Y. She, S. Li, B. Du, et al., On-orbit assembly mission planning considering topological constraint and attitude disturbance, *Acta Astronaut.* 152 (2018) 692–704.
- [108] A. Calvi, Spacecraft structural dynamics & loads an overview, *Satellite Engineering Class* (2010).
- [109] Y. Li, J. Onoda, K. Minesugi, Simultaneous optimization of piezoelectric actuator placement and feedback for vibration suppression, *Acta Astronaut.* 50 (6) (2002) 335–341.
- [110] S.S. Narayan, P.S. Nair, A. Ghosal, Dynamic interaction of rotating momentum wheels with spacecraft elements, *J. Sound Vib.* 315 (4–5) (2008) 970–984.
- [111] K. Komatsu, H. Uchida, Microvibration in spacecraft, *Mechanical Engineering Reviews* 1 (2) (2014). SE0010-SE0010.
- [112] I. Chopra, J. Sirohi, *Smart Structures Theory*, Cambridge University Press, 2013.
- [113] Xin Guan, *Integrated Design of Vibration Isolation and Attitude Control for High Resolution Remote Sensing Satellites*, Tsinghua University, 2012.
- [114] Yang Lin, *Large Swath and High Resolution Satellite Integrated Structure Multi-Objective Optimization Design*, University of Chinese Academy of Sciences (Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences), 2019.
- [115] J.K. Wayer, J.F. Castet, J.H. Saleh, Spacecraft attitude control subsystem: reliability, multi-state analyses, and comparative failure behavior in LEO and GEO, *Acta Astronaut.* 85 (2013) 83–92.
- [116] Tim Hindle, Torey Davis, Jim Fischer, Isolation, pointing, and suppression (IPS) system for high-performance spacecraft, in: Proc. SPIE 6527, Industrial and Commercial Applications of Smart Structures Technologies 2007, 11 April 2007, p. 652705, <https://doi.org/10.1117/12.720663>.
- [117] Lei Liu, Benli Wang, Development of Stewart platforms for active vibration isolation and precision pointing, in: Proc. SPIE 6423, International Conference on Smart Materials and Nanotechnology in Engineering, 31 October 2007, p. 642347, <https://doi.org/10.1117/12.779866>.
- [118] Anderson E H, Leo D J, Holcomb M D. Ultraquiet platform for active vibration isolation[C]//Smart Structures and Materials 1996: Smart Structures and Integrated Systems. International Society for Optics and Photonics, vol. 1996, 2717: 436–451.
- [119] T. Wacker, L. Weimer, K. Eckert, GOCE platform micro-vibration verification by test and analysis, in: Proceedings of the European Conference on Spacecraft Structures, Materials and Mechanical Testing, 2005. ESA SP-581). 10–12 May 2005, Noordwijk, Netherlands.
- [120] A. Grillenbeck, IABG mbH, Germany Ottobrunn, M. Wagner, ESA/ESTEC, Micro-Vibration qualification on spacecraft structures: a challenge for test Engineers, Manhattan Beach, Calif, oct12- 15 (2009).
- [121] N. Jedrich, D. Zimbelman, M. Turczyn, J. Sills, C. Voorhees, B. Clapp, Cryo cooler induced micro-vibration disturbances to the Hubble space telescope, in: 5th Cranfield Space Dynamics Conference., July 2002.
- [122] A. García-Pérez, A. Ravanbakhsh, F. Sorribes-Palmer, et al., Structural shock verification by numerical analysis of the EPD payload units on board Solar Orbiter spacecraft, *Acta Astronaut.* 168 (2020) 282–292.
- [123] G. Park, A.C. Rutherford, J.R. Wait, et al., High-frequency response functions for composite plate monitoring with ultrasonic validation, *AIAA J.* 43 (11) (2005) 2431–2437.

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